

HDV/HDE Booking Tool

Business Requirement Document — v6

Version: 6.0
Date: 2026-04-17
Based on: BRD v5 (2026-04-15) + Requirements gathered in meeting 16.04.2026
Scope: Phase 1 unless noted as Phase 2
Prepared by: Telman Nurzhanov (SA)

Changelog v5 → v6

#	Раздел	Изменение
1	3. Glossary	Work Center — добавлено определение и таблица из 15 примеров кодов шагов (LRG, MGT, BHOE, DTRK и др.); список будет скорректирован Асылбеком
2	3. Glossary	Уточнено соотношение Work Center ↔ Equipment: один тип Equipment работает только в одном WC; один WC может включать несколько типов Equipment
3	5.1 Equipment	Убрано поле «Операционный статус» из параметров карточки
4	5.1 Equipment	Добавлены явные поля карточки: Заморозка (флаг), Причина заморозки , Дата завершения заморозки
5	5.1 Equipment	Добавлено поле Принадлежность (TCO Owned / Long term rented) в блок параметров карточки
6	5.1 Equipment	Добавлена политика: максимальное количество полей карточки выносятся в справочники (настраиваются через Admin Panel), в т.ч. классы, категории, подкатегории, производитель, модель, компании, локации, рабочие центры, дивизионы, группы, департаменты, отделы
7	5.3.2 / 8.1	Обновлена таблица требуемых полей WO из DataLake: добавлены Описание статуса , Объёмы шагов (требуемое количество

#	Раздел	Изменение
		единиц техники), Диапазон дат (дата начала / завершения)
8	5.3.1 / 6.4	Уточнена логика поля WO/WC для Логистики : поля Work Order и Work Center — необязательные (Логистика не работает в JDE и не имеет Work Center)
9	6.4 Request Management	Добавлен FR-NEW-48 : на странице «Подать заявку» — блок «Выбрать Work Order из JDE»; система отображает список WO с атрибутами; выбор WO → система формирует draft заявки; пользователь дозаполняет технику и время
10	6.3 Equipment Management	Добавлен FR-NEW-49 : поля «Заморозка», «Причина заморозки», «Дата завершения заморозки» — явные поля карточки; видны FO и Admin
11	10. Open Questions	OQ-NEW-G —  CLOSED : Work Center = тип работ (шаг), не атрибут техники. Один тип Equipment → один WC; один WC → несколько типов Equipment
12	10. Open Questions	Добавлен OQ-NEW-I : Приоритет — атрибут заявки, шага в заявке или техники в шаге заявки?
13	10. Open Questions	Добавлен OQ-NEW-J : Департамент в отчётах — по Fleet Owner или по Equipment? (могут принадлежать разным департаментам)
14	11. New Requirements	Добавлены FR-NEW-48, FR-NEW-49 в сводную таблицу

Status Legend

Icon	Meaning
	Confirmed — requirement confirmed in meetings, no conflicts
	Partially confirmed / open question — requirement accepted but details TBD
	Obsolete / Superseded — cancelled or replaced by another requirement
	New — not in original BRD; identified during requirements gathering

Icon	Meaning
	Conflict — contradiction between BRD and meeting decisions; requires explicit sign-off
	Phase 2 — confirmed in scope, deferred to Phase 2
	Backlog — confirmed direction, deferred after baseline implementation

1 Introduction

1.1 Purpose

This document defines the business requirements for a booking tool that allows users to create requests for HDV/HDE equipment. Requests may include multiple pieces of equipment, each of which is treated as a separate booking with its own approval workflow.

1.2 Scope

The tool will support bookings for company-owned equipment as well as equipment provided by business partners. It will cover request creation, approval flow, status tracking, reporting, and integration with external systems where applicable.

Scope decision (08.04): Booking through the tool is available only for TCO Owned and Long-term rented equipment. On-demand BP (external fleet) is displayed as a catalog/showcase only — no booking workflow for on-demand BP in the system. This is a significant scope reduction vs. original BRD. See Section 6.7 for detail.

2 Business Context

2.1 Current Challenges

- Lack of centralized booking and collaboration between departments on shared resources
- Low transparency on total number of units and spend
- No standardized process or guidelines
- No Process Owner(s)
- Lack of visibility due to manual utilization monitoring

2.2 Business Objectives

- Standardize the request and booking process.
- Ensure automated routing of approval based on equipment fleet ownership.
- Provide transparency and accountability in the booking process.
- Enable scalability for both internal and partner-provided equipment.
- Process ownership by Logistics.

3 Glossary

Request – a container grouping multiple equipment bookings (one per equipment item).

Regular Request – a request created by Requestor in HDV/HDE booking tool.

Service Work Request – a request created in HDV/HDE automatically via API based on Work Order created by Requestor in JDE E1 system.

Booking – individual booking within a request. One booking is linked to one unique equipment item. Treated individually within a request.

Equipment – individual asset (HDV/HDE unit) that can be booked through the system. Each equipment item belongs to a fleet and inherits approval responsibility from the corresponding Fleet Owner.

Fleet – logical group of equipment owned by a Fleet Owner (for internal: Maintenance, Construction, TFM, etc.; for external: business partner's name). Each fleet has a unique name assigned at creation.

Fleet Owner (FO) – person responsible for approvals within a fleet. One fleet can have 2+ Fleet Owners (e.g. shift workers). A fleet has a shared team email used for system notifications.

Approver – Fleet Owner and/or Cost Center (CC) owner, who confirms/declines bookings.

 **Mobilization** – the period of preparation/movement of equipment before it arrives at the work site. Booking period starts from mobilization start, not from arrival. (*Мобилизация — период подготовки/перемещения техники до прибытия на объект*)

 **Equipment categories (Usage Status):**

- **Assigned** – equipment assigned to a specific department; visible to all users, bookable only by authorized users with FO role. (*Закреплённая техника*)

- **Shared without Conditions** – equipment available for general booking without any preconditions. (*Общедоступная без условий*)
- **Shared with Conditions** – equipment available for booking, but the Requestor must provide a **Justification** field. Used for specialized equipment where FO needs context (e.g. specific compressors, chemical tanks). Fleet Owner can recall any booking at any time regardless of this status — recall is a separate mechanism. (*Общедоступная с условиями: требуется обоснование. [Updated 13.04 — OQ-NEW-D CLOSED]*)

NEW Equipment Ownership Types:

- **TCO Owned** – internally owned by TCO.
- **Long-term rented** – rented long-term; treated as internal fleet for booking purposes. Managed by internal TCO team (not by BP).
- **On-demand rented (BP Showcase)** – rented on demand; displayed as catalog/showcase only in Phase 1.

NEW Utilization (*Утилизация*) – daily efficiency of equipment usage; reflects how effectively the company manages its equipment.

- Tracked by **mileage** (for wheeled equipment) or **engine hours** (for other equipment types), or **both** depending on equipment category.
- Each Fleet Owner independently defines the **100% utilization threshold** for each equipment type under their fleet (maximum moto-hours or km per 24 hours = 100%).
- **Maximum Utilization Capacity per Shift** — the maximum possible utilization of a given equipment unit within a working shift; serves as the 100% reference point for utilization calculations. (*Максимальная возможная утилизация в рабочую смену*)
- Utilization percentage is calculated per day, month, or any selected period.
- **NEW [Updated 13.04]:** Some equipment types have **both** mileage and engine hours parameters. Planned values are entered by FO; actual values are sourced from trackers only (not manually editable).

NEW [Updated 14.04] Utilization Calculation Formulas:

Metric type	Formula
Engine hours only	$\text{utilization} = \text{engine_time} / \text{target_mh_per_day}$

Metric type	Formula
Mileage only	<code>utilization = mileage / target_km_per_day</code>
Both (hours + km)	<code>utilization = max(engine_time / target_mh_per_day, mileage / target_km_per_day)</code>
Cap rule	If <code>utilization > 1</code> → display as 1 (100%)

Business goal of utilization monitoring (14.04): Eliminate equipment downtime (idle time). Primary metric — how effectively equipment is used. Exceeding 100% is capped at display level; the actual value does not block booking unless a threshold is configured (see OQ-NEW-H).

NEW [Updated 13.04] Criticality (*Критичность*) — equipment marked as critical requires **security escort (service de sécurité)** during transportation. Not a maintenance priority indicator. Displayed in equipment card.

NEW [New 16.04] Work Center (*Рабочий центр / Шаг*) — объём работ, которые нужно выполнить в рамках Work Order; называется «шагом», так как работы выполняются последовательно. Является атрибутом Work Order Step, а не атрибутом единицы техники.

- Ключевые ограничения:**
- Один тип Equipment работает только по **одному** Work Center.
 - Один Work Center может обслуживаться **несколькими** типами Equipment.

Примеры кодов Work Center (предварительный список — подлежит корректировке Асылбеком):

Код	Описание
LRG	Группа ГПМ
MGT	Механики по дизелям
BHOE	Экскаватор, трактор ОТО
DTRK	Самосвал ОТО
FTRK	Грузовик ОТО

Код	Описание
GDRV	Грузовик с краном ОТО
HYDR	Водоструйная очистка ОТО
OILT	Oil Tanker
VTRK	Вакуумная машина ОТО
CDECK	Транспортная платформа ОТО
CRANE	Кран ОТО
FORKL	Вилочный погрузчик ОТО
GMLOG	Механик передвижного оборудования
MLIFT	Подъёмник ОТО
MSINS	ППУ

Note: Logistics subdivision does not operate Work Centers in JDE. WO and WC fields are optional for Logistics. (OQ-NEW-G CLOSED 16.04)

4 Stakeholder & Roles

4.1 Requestor

All users in TCO network will by default have Requestor role to book equipments of the internal fleet (TCO owned + Long-term rented HDV/HDE). On-demand external (BP) equipment is displayed as a showcase/catalog only — booking of external equipment is done in DCT, not in this tool.

Requestors are allowed to:

1. Create a Regular Request — 
2. Submit the Regular Request — 
3. Prolong booking — 
4. Submit feedback on the booked equipment — 
5. Monitor utilization of the booked equipment for the booking period, if the equipment has tracker(s) — 

6. Terminate booking (only external fleet) — ❌ (external fleet booking is not available in the system)
7. NEW Close booking by pressing "Close" button (manual close only) — ✅ (Закрытие брони по кнопке)
8. NEW Book Assigned equipment only if authorized by Administrator — ✅ (Бронирование закреплённой техники — только при авторизации Admin)

4.2 Service Work Processor

⚠️ **PARTIALLY CLARIFIED (13.04):** Based on meeting discussion, the Service Work Processor role is likely **graphists / schedulers** who work with Work Orders in JDE and process drafts once a Work Order step reaches a confirmed status (≈55 in JDE). They push Service Work Requests from Draft to Submitted for Fleet Owner processing. Formal confirmation from TCO required (OQ-NEW-C).

NEW **Action (14.04):** Schedule a dedicated meeting with graphists — **Асылбек, Диас** — to confirm role scope, specific functions and permissions.

~~Access provision is managed by Azure Active Directory (AAD) group. Similar to Requestor, however, Service Work Processor works with Service Work Requests, created automatically in HDV/HDE booking tool.~~

4.3 Fleet Owner

For internal Fleet Owners access provision is managed by AAD group. There can be 2+ AAD groups to manage internal Fleet Owners (separate group for each fleet created by Administrator).

NEW **[Updated 13.04]:** One fleet can have **two or more Fleet Owners** (e.g. shift workers / сменщики). Both owners are displayed in the equipment card. Each fleet has a **shared team email** address used as a single notification target (alongside personal emails via AD group). When a Fleet Owner changes role, Administrator reassigns the fleet without rebuilding the fleet structure.

Fleet Owners are allowed to:

1. NEW **[Updated 10.04]** Create new equipment under own fleet (allowed for **both internal and external** Fleet Owners) — ✅ for internal; 🟦 Phase 2 for external fleet onboarding workflow
2. Update equipments (only parameters that are allowed for editing by Administrator) — ✅
3. Delete equipment (soft delete) — ✅
4. Confirm / decline requests to book equipments under own fleet — ✅

5. NEW **[Updated 10.04 — CONFLICT-01 RESOLVED]** Update booking period **before or after confirmation**; Requestor is notified automatically; no re-confirmation required — ✔
6. Change equipment in booking — ✔
7. Terminate already confirmed bookings (only internal fleet) — ✔
8. Freeze / unfreeze own equipment from booking for a period or without end date — ✔
9. Monitor utilization of own equipment, if the equipment has tracker(s) — ✔
10. NEW Press "Mobilization started" button to fix mobilization start time — ✔ (*Кнопка «Мобилизация началась»*)
11. NEW Delegate FO privileges to any TCO employee for a specified period (without IT involvement) — ⚠ Pending cybersecurity sign-off (OQ-NEW-E)
12. NEW One user can be an owner of several fleets (member of several AAD groups) — ✔
13. NEW **[Updated 13.04]** One fleet can have multiple Fleet Owners (shift workers); both displayed in equipment card with shared team email — ✔ (*OQ-30 partially closed*)

4.4 CC Owner (incl. DOA)

[Confirmed 10.04]: CC Owner confirmation is NOT required for Long-term rented equipment — this follows the internal booking flow. CC Owner is applicable only to on-demand external fleet booking, which is in DCT system.

CC owners are allowed to:

1. Confirm / decline external fleet equipment booking request — ✗ **Obsolete** (*CC Owner approval applies only to on-demand external fleet booking, which is done in DCT — not in this tool*)

4.5 Administrator

Administrator role will be provided to process owners – Logistics. Access provision is managed by Azure Active Directory (group). Users with an Administrator role are allowed to:

1. Create/edit internal fleet and link with AAD group — ✔
2. Create/edit external fleet and manage external fleet owners (send invitation, deactivate external user's account) — ● Phase 2
3. Delete fleet — ✔
4. Create internal fleet equipment — ✔
5. Update equipment's parameters (extended edit) — ✔
6. Delete equipment (soft delete) — ✔
7. Manage 'Request' template — ⚠ Partially clarified (OQ-NEW-B); **baseline fields first; custom attribute management is backlog**

8. Manage 'Equipment' template — ⚠️ Partially clarified (OQ-NEW-B); **adding custom attributes (columns/filters) per equipment type — backlog after baseline fields agreed**
9. Manage 'Booking' template — ⚠️ TBD (OQ-NEW-B)
10. Access to custom reportings — ✅
11. Access to utilization dashboard(s), if the equipment has tracker(s) — ✅
12. **NEW** Manage all configurable system parameters (booking horizon, max duration, FO timeout thresholds, etc.) via Admin Panel — ✅
13. **NEW** Authorize specific users to book Assigned equipment — ✅
14. **NEW** Manage delegation settings — ⚠️ Pending cybersecurity sign-off (OQ-NEW-E)
15. **NEW** **[Updated 13.04]** Manage fleet shared team email — ✅ (*Управление shared email флота*)
16. **NEW** **[Updated 16.04]** Manage reference/handbook values (equipment classes, categories, subcategories, manufacturers, models, companies, locations, work centers, divisions, groups, departments, units) via Admin Panel — ✅

4.6 **NEW** Transportation Responsible (Мобилизация/Транспортировка)

A dedicated system role for the team responsible for transporting non-motorized (Unwheeled) equipment (Heavy Ops / Maintenance team).

This role is allowed to:

1. **NEW** Receive notification about transportation request for Unwheeled equipment — ✅
2. **NEW** Confirm or decline the transportation request — ✅

5 Entities

5.1 Equipment

All equipments must have a mandatory attribute – fleet. The attribute will affect on which template the equipment will have (depending on internal and external fleet), who can create/edit/delete equipments (fleet owners), approval and processing flow of the equipment booking. — ✅

NEW **Fleet naming (10.04):** Each fleet must have a unique **name** assigned at creation (e.g., "Heavy Ops", "Logistics", "Maintenance", "Production"). Equipment assignment is linked to a specific employee (not a department); when a Fleet Owner changes, the Administrator manually re-assigns the fleet to the new person.

NEW Fleet Owners display (13.04): Equipment card displays **up to 2+ Fleet Owners** (shift workers) alongside their **shared team email**. Notifications are sent to shared team email and/or personal emails via AD group.

NEW [New 16.04] Work Center assignment: Each equipment type is assigned to **exactly one Work Center**. This assignment is a system parameter managed via the Admin Panel reference. The work center defines what type of work the equipment performs within a Work Order step.

NEW [New 16.04] Reference/Handbook policy: The maximum number of equipment card fields must be configured via system handbooks (reference data), editable through the Admin Panel. The following are handbook-managed:

- Equipment classes (HDE, HDV)
- Categories and subcategories
- Manufacturer, Model
- Companies (TCO White Pages)
- Locations
- Work Centers
- Divisions, Groups, Departments, Units (Отделы)

Internal fleet equipments have the next parameters:

1. Basic fixed info (TCO ID, Description, Service Zone, Class/Category, License Plate *(if present — not all equipment has one)*, Serial number, etc.) —   (Базовый перечень полей разбирается по типам техники — OQ-NEW-A частично закрыт)
2. Dynamic (changeable) parameters: Equipment Usage Status, **Cost Center**, CC/Division/Group/Department/Unit, Assigned location, Maintenance BP, Contact number / email, etc. — 

NEW [Updated 16.04]: «Операционный статус» (Operational Status) **removed** from equipment card parameters.

3. **NEW [New 16.04] Ownership field** — «Принадлежность» with values: **TCO Owned / Long term rented** — 
4. **NEW [New 16.04] Freeze fields** in equipment card:
 - **Заморозка** (freeze flag: frozen / not frozen)
 - **Причина заморозки** (freeze reason — free text or reference value)
 - **Дата завершения заморозки** (freeze end date; empty = frozen indefinitely)

These fields are visible and editable by FO and Admin. They reflect the state managed via FR-021.

5. Equipment specifications (picture, Chassis type, Drive type, Loading capacity, Generator / Compressor specifications) —
6. Trackers' IDs if exist (WIALON tracker "Group", IVMS tracker "Registration number", Vega tracker "Device ID", etc) — **[Updated 13.04]: Multiple trackers per unit supported** (e.g. one for engine hours, one for GPS/mileage)
7. **[Updated 13.04]** Planned engine hours (per shift/day) — editable by FO —
8. **[Updated 13.04]** Planned daily mileage (km/day) — editable by FO —
9. **[Updated 13.04]** Actual engine hours / mileage — sourced from tracker only; not manually editable —

Equipment card UI additions (13.04):

- **Booking calendar:** visual calendar in the equipment card (FO view) showing which periods are booked; confirmed required
- **Criticality flag:** displayed in card; meaning = security escort required during transport

[Updated 14.04] Equipment card UI design details:

- **Work schedule display:** split into **day shift / night shift** blocks
- **Weekend coverage:** work schedules account for weekend days — default shift 06:00–06:00; for Long Term equipment — 07:00–07:00
- **Multiple photos:** equipment card supports uploading **multiple photos** per unit (*confirmed 14.04*)
- **Sidebar:** Comments block placed in sidebar panel
- **Tracker block:** includes the ability to **add new trackers** from the equipment card UI (not only view existing)
- **[New 16.04] Schedule slider:** slider control added to the work schedule block for selecting time ranges

License plate and TCO number rules (13.04):

- **License plate (роchomep):** present on most wheeled equipment; may be absent on some stationary units. Field optional/conditional.
- **TCO number (TШO-номер):** assigned when equipment is purchased for TCO; long-term rented equipment arrives with its own license plate and may not have a TCO number. Business partners (L-rent) may have neither.

Equipment Classification (three dimensions):

Dimension	Values
Mobility	Self-propelled / Not self-propelled (Unwheeled) / Stationary
Ownership	TCO Owned / Long-term rented / On-demand rented (BP Showcase)
Usage Status	Assigned / Shared without Conditions / Shared with Conditions

- Note (08.04): Mobility category is NOT displayed in Requestor UI — Requestor sees a flat list of equipment types. ✓
- Note (08.04): Stationary HDE is excluded from Requestor search — stored in system for FO use only. ✓
- Note (13.04 — OQ-NEW-D CLOSED): Shared with Conditions differs from without Conditions only in the presence of a mandatory **Justification** field upon booking. FO can recall any booking at any time — this is a separate mechanism independent of conditions type. ✓

NEW

Equipment availability display rules:

- Equipment under maintenance is shown grayed out, not selectable; label "Under maintenance until [Planned date from JDE]" ✓
- Repair status is displayed in equipment list as well as in equipment card ✓
- Repair status and planned completion date are sourced from JDE → DataLake integration — ⚠ (depends on integration)

NEW

5.1.1 Equipment Type-Specific Parameters (13.04)

The parameters below define specific attributes and search filters for each equipment type. These are used both in the equipment card and in the Requestor search/filter form. All parameters are preliminary and subject to final sign-off from TCO Fleet Owners. Remaining types to be covered per OQ-NEW-F.

Loader / Погрузчик

Parameter	Values / Format	Used in filter
Load capacity (грузоподъемность)	Ranges: up to 2 t / 2–4.5 t / 4.5–8 t	✓ Dropdown or range
Type	Fork (вилочный) / Bucket (ковшовый)	✓

Parameter	Values / Format	Used in filter
Engine type	Electric / Diesel	

Crane-Manipulator / Кранманипулятор

Parameter	Values / Format	Used in filter
Vehicle load capacity	numeric (tons)	
Crane load capacity	numeric (tons)	
Boom length	numeric (meters)	
Body length (кузов)	numeric (meters)	

Mobile Generator / Генератор передвижной

Parameter	Values / Format	Used in filter
Power (мощность)	Dropdown: 30 / 60 / 80 / 100 / 120 / 130 / 200 / 250 kW (exact list TBD per fleet data)	Checkbox multi-select
Voltage (напряжение)	—	TBD
Current (ток)	—	TBD
Battery capacity (ёмкость батарей)	—	TBD
Mobility type	Wheeled / Non-wheeled / Stationary	

Action (Асылбек): provide final grouping of generator power categories from master fleet file.

Motor Compressor / Компрессор моторный

Parameter	Values / Format	Used in filter
Pressure (давление)	numeric (atm); dropdown or range TBD	
Productivity (производительность)	numeric (m³/h); dropdown or range TBD	

Action (Асылбек): provide grouping of compressor pressure and productivity categories.

Dump Truck / Самосвал

Parameter	Values / Format	Used in filter
Load capacity (тоннаж)	Dropdown or range (exact ranges TBD)	☒ Primary filter
Body type (тип кузова)	Open / Closed (tent) — secondary; user can also reference photo	Via photo / optional

Frac Tank / Фрактанк

Parameter	Values / Format	Notes
No type-specific filters	All frac tanks have the same volume	Any available unit can be selected
Dedicated transport vehicle	Kenworth Winch Truck (1 unit in fleet)	System note: if transport vehicle is unavailable, frac tank cannot be deployed

Decision point: transport vehicle for frac tank is the only Kenworth Winch Truck. System behavior when it is occupied — TBD (not currently part of booking workflow).

Cargo Trailer / Прицеп грузовой трейлер карго

Parameter	Values / Format	Used in filter
Type	Low-frame (низкорамный) / Standard	☒
Load capacity	numeric (tons)	☒
Suspension system	Pneumatic / Standard	☒

Semi-truck / Тягач HD

Parameter	Values / Format	Notes
No independent filter	Tractor unit is selected by FO based on trailer type	Requestor does not select tractor independently

Decision (13.04): Tractor head is determined by the trailer that needs to be towed. FO selects appropriate tractor. Requestor-facing filter for semi-truck is not required.

5.2 Booking

Booking is linked to a single equipment and have booking range (datetime). Several bookings form a single request. — ✓

NEW Architecture decision (09.04): Data model follows Order/OrderLine pattern: one Request contains multiple RequestItems (one per equipment); each RequestItem independently processed by FO and generates a separate Booking. ✓

5.2.1 Internal Booking

Booking of TCO owned (internal fleet, including Long-term rented) equipment. Internal Fleet equipments require only confirmation from the side of Internal Fleet owner. Fleet owner can apply adjustments to the booking period and/or replace equipment if substitution exists. — ✓

NEW Mobilization (09.04): Booking period starts from mobilization start, not from arrival at work site. FO presses "Mobilization started" button to fix the actual start in the system. ✓

NEW [Updated 10.04 — CONFLICT-01 RESOLVED]: FO can adjust the booking period **both before and after confirmation**. Requestor is notified automatically upon any change. No re-confirmation from Requestor is required. ✓

5.2.2 External Booking

! CONFLICT / PHASE CHANGE vs. original BRD:

Original BRD described a two-step approval flow (CC Owner → External FO) for external fleet equipment. **Meeting decision (08.04): On-demand BP equipment is NOT bookable through the system in Phase 1.** It is displayed as a catalog/showcase only.

- On-demand rented (BP Showcase) — catalog view only; no booking workflow in Phase 1. ✓
- Long-term rented — treated as **internal fleet**; no CC Owner step required. ✓ (*Confirmed 10.04*)
- Full external booking workflow (CC Owner → External FO → DCT/RWA) — ✗ **Obsolete**

5.2.3 **NEW** Booking Snapshot

At the moment of booking confirmation, the system saves a snapshot of key equipment attributes (model, license plate, fleet name). This ensures historical readability when equipment data is later updated. — ⚠ Preliminary (OQ-37)

5.3 Request

Requests have 2 types: Regular Request, Service Work Request. The type affects on request template, mandatory of filling parameters and approvers' workflow. — 

5.3.1 Regular Request

In a Regular Request Requestor can add **1 or more** equipments from any fleet; there is no restriction that all equipments must be from the same fleet. Each added equipment will be considered as separate booking and will be confirmed by Fleet Owners separately. — 

[Updated 10.04]: Minimum equipment count in a request is **1** (not 2+).

The system will prioritize TCO owned (internal fleet) assets as a **business rule** (not technically enforced at current selection model). If FO does not respond within timeout — Requestor is redirected to the BP catalog.

Request fields (08.04):

- **Work Order (JDE)** — mandatory for Maintenance / Railroad / Operations fleets. One Work Order per request. 
- **Work Center (WC)** — mandatory for fleets that operate in JDE; **not applicable to Logistics** (*Logistics does not work in JDE — WC field is optional/hidden for Logistics*). 
- **Location** (text input) — replaces Work Order field for SCM Logistics fleet. 
- **Work Description** — mandatory, ~50 character limit. 
- **Comments** — optional, no hard limit. 
- **Priority (P1–P4)** — selected by Requestor.  (*Level of attribution — request vs. step vs. equipment — see OQ-NEW-I*)

Note (10.04): Within one request (Work Order), all equipment items share the same **location**, but may have **different booking time periods**.

 **[New 16.04] WO pre-fill flow:** On the "Submit Request" page, a dedicated block **"Выбрать Work Order из JDE"** allows Requestor to select an existing WO from JDE. The system retrieves the WO list from DataLake (see Section 8.1 for field set) and upon WO selection **automatically generates a draft request** pre-populated with WO attributes (number, name, priority, steps with volumes). Requestor then fills in remaining fields (equipment selection, booking time). See FR-NEW-48.

Request constraints (03.04):

- Maximum booking horizon: 1 month ahead (configurable by Admin). 

- Maximum single booking duration: 1 week (configurable by Admin). 

 **Form** → **Bookings split (09.04)**: Requestor fills a single form for a Work Order with multiple equipment items. Upon submission, the system **automatically splits** the form into atomic individual bookings — one per equipment item. All bookings from one form are linked via the Work Order field. 

5.3.2 Service Work Request (within Work Order in JDE E1)

 **PARTIALLY CLARIFIED (13.04)**: Service Work Requests are auto-created as Drafts in the tool when a Work Order step reaches a specific status in JDE. Based on discussion, Work Orders progress through statuses: ~50 (scheduler review) → **55 (released to operations)**. Steps at status 55 are candidates for import. Service Work Processor (likely a scheduler/graphist) reviews the draft and submits it to Fleet Owner. Each 'Step' in Work Order will be created as a separate Service Work Request. Formal confirmation from TCO required (OQ-NEW-C).

 **[Updated 16.04] Required WO data fields from DataLake** (expanded from v5):

Field	Description / Notes
Work Order number	Unique identifier in JDE E1
Work Order name	Need examples from TCO
Status	Clarify list of valid statuses in JDE
Status description	 [New 16.04] Text description of the WO status
Priority	Priority value in JDE E1
Work Center (Step code)	Type of work; e.g. "CRANE", "DTRK"; FO uses step to identify equipment type needed
Step name	Text name of the Work Center step
Step volumes	 [New 16.04] Required quantity of equipment units per step
Date range	 [New 16.04] Start date / End date of the WO step

 **[Updated 14.04] Data source clarification**: WO data is extracted from **Data Lake**, not directly from JDE API. Data Lake sync frequency: **every 30 minutes**. Action: verify exact fields available in Data Lake for WO (Аян, Тельман).

6 Functional Requirements

6.1 User & Role Management

ID	Requirement	Status
FR-001	System supports role-based access control	✅ Confirmed
FR-002	All users in TCO network have access to the tool as a Requestor; to add external fleet equipment, the user must be a member of "RWA initiator" AAD group	❌ Obsolete — booking of external fleet equipment is in DCT, not this tool
FR-003	Administrators, Service Work Processors are defined by membership in the corresponding AAD groups	⚠️ SWP pending (OQ-NEW-C)
FR-004	The system will define if the user has CC Owner or DOA privileges as a result of integration with PSWS and DOA	❌ Obsolete — CC Owner role is not applicable; external fleet booking is in DCT
 FR-NEW-01	All time-based parameters (FO response timeout, booking horizon, max booking duration) are configurable via Admin Panel and are NOT hardcoded	✅ Confirmed
 FR-NEW-02	FO can delegate own privileges to any TCO employee for a specified date range (start + end date); delegation is automatically revoked upon expiry; delegation scope is configurable	⚠️ Pending cybersecurity sign-off (OQ-NEW-E)

6.2 Admin Panel

ID	Requirement	Status
FR-005	Admin can create fleets (internal, external); each fleet must be given a unique name	✅ Confirmed
FR-006	Admin can link internal fleet with AAD group, so that members of that group can act as Fleet Owners	✅ Confirmed

ID	Requirement	Status
FR-007	Admin can manage owners of created external fleets: send invitation to BP, deactivate existing external accounts	🟦 Phase 2
FR-008	Admin can update existing fleets	✅ Confirmed
FR-009	Admin can delete existing fleets	✅ Confirmed
FR-010	Admin can manage template of equipment (internal and external): define parameters, formats, mandatority, editability by owners. [Updated 13.04] Baseline fields defined first; ability to add custom attributes (new columns/filters per equipment type) is backlog	⚠️ Baseline first; custom attributes → 📋 Backlog
FR-011	Admin can manage template of request. [Updated 13.04] Backlog after baseline fields agreed	📋 Backlog (OQ-NEW-B)
FR-012	Admin can manage template of booking	⚠️ TBD (OQ-NEW-B)
FR-013	Admin can create equipments of internal fleet	✅ Confirmed
FR-014	Admin can edit (extended edit) equipments	✅ Confirmed
FR-015	Admin can delete equipments	✅ Confirmed
FR-016	Admin has access to all dashboards and reports	✅ Confirmed
 FR-NEW-03	Admin can configure all system time parameters: FO response timeout (default 24h reminder / 48h escalation), max booking horizon, max booking duration	✅ Confirmed
 FR-NEW-04	Admin can authorize specific users (by employee ID or role) to book Assigned equipment and configure justification requirement	✅ Confirmed
 FR-NEW-42	[New 13.04] Admin can set and update shared team email for each fleet; used as notification target alongside personal emails via AD group	✅ Confirmed
 FR-NEW-50	[New 16.04] Admin can manage all reference/handbook values via Admin Panel: equipment classes, categories, subcategories,	✅ Confirmed

ID	Requirement	Status
	manufacturers, models, companies (TCO White Pages), locations, work centers, divisions, groups, departments, units	

6.3 Equipment Management

ID	Requirement	Status
FR-017	[Updated 10.04] Both internal and external Fleet Owners can create equipment under their own fleet. Long-term rented equipment is created by the internal TCO team (not by BP).	✓ Confirmed
FR-018	Fleet owner must indicate 'Fleet' of the equipment; system allows selecting only fleets for which the user is a FO	✓ Confirmed
FR-019	Fleet owner can edit (allowed by Admin) equipments under his fleet; one user can be an owner of several fleets	✓ Confirmed
FR-020	Fleet owner can delete equipments under his fleet	✓ Confirmed
FR-021	Fleet owner can freeze / unfreeze own equipments for a period of time or without end date	✓ Confirmed
FR-022	Requestor / Service Work Processor can submit feedback on equipment with confirmed booking, starting from the booking start datetime	✓ Confirmed
 FR-NEW-05	FO can upload multiple photos for equipment; Requestor sees equipment photo(s) in the request form	✓ Confirmed (<i>multi-photo confirmed 14.04</i>)
 FR-NEW-06	Equipment repair status (maintenance in progress / completion date) is sourced from JDE → DataLake integration and displayed in Requestor search results and equipment list	⚠ Pending (JDE integration detail)
 FR-NEW-07	Stationary HDE equipment is excluded from Requestor search results; stored in system for FO use only	✓ Confirmed
 FR-NEW-08	Assigned equipment is visible to all users in search, but can only be booked by users authorized by Admin; "Justification" field is mandatory for Assigned equipment booking	✓ Confirmed

ID	Requirement	Status
 FR-NEW-09	Shared with Conditions equipment is displayed with a visual marker (color coding)	 Confirmed
 FR-NEW-40	[New 13.04] Equipment card (FO view) displays a booking calendar — visual representation of booked periods (e.g. color-coded days) to help FO plan maintenance and availability	 Confirmed
 FR-NEW-41	[Updated 14.04] An equipment unit can have multiple trackers linked simultaneously (e.g. engine hours tracker + GPS/mileage tracker); all tracker IDs stored in equipment card; FO can add new trackers from the equipment card UI	 Confirmed
 FR-NEW-49	[New 16.04] Equipment card contains explicit freeze fields: Заморозка (flag), Причина заморозки (reason), Дата завершения заморозки (end date; empty = indefinite freeze). Visible and editable by FO and Admin. Freeze/unfreeze action (FR-021) writes to these fields.	 Confirmed

6.4 Request Management

ID	Requirement	Status
FR-023	Requestor can create a Request	 Confirmed
FR-024	Request has a unique identifier and metadata (date, requester, status, etc.)	 Confirmed
FR-025	Requester can view request details and status	 Confirmed
FR-026	Request can have status: Draft, Submitted, In Progress, Completed	 Aggregated statuses TBD (OQ-36)
FR-027	Requester can edit or cancel a draft request before submission	 Confirmed

ID	Requirement	Status
FR-028	Requests created by system automatically (Service Work Requests) will have 'Draft' status before submission by Service Work Processor	⚠ Pending (OQ-NEW-C)
FR-029	Only users with role Service Work Processor can manage (submit) Service Work Requests	⚠ Pending (OQ-NEW-C)
FR-030	System supports draft saving (continue later)	✅ Confirmed
 FR-NEW-10	TO-BE: only on-demand booking; weekly schedule management is out of scope	✅ Confirmed
 FR-NEW-11	Requestor selects a booking priority P1–P4 at the time of request creation	✅ Confirmed (<i>Priority level attribution — see OQ-NEW-I</i>)
 FR-NEW-12	[Updated 16.04] Work Order field (JDE) is mandatory for Maintenance/Railroad/Operations fleets. For Logistics subdivision: WO and WC fields are optional (Logistics does not operate in JDE). For SCM Logistics: Location (free text) replaces WO. Work Order is one-to-one with a request.	✅ Confirmed
 FR-NEW-13	"Work Description" field is mandatory in the request form (~50 character limit)	✅ Confirmed
 FR-NEW-14	"Comments" field is optional in the request form (no hard character limit)	✅ Confirmed
 FR-NEW-15	Requestor submits a single form for a Work Order with multiple equipment items; the system automatically splits the form into individual bookings (one per equipment item); all bookings are linked via the Work Order field	✅ Confirmed

ID	Requirement	Status
 FR-NEW-16	Partial confirmation: confirmed items within a request proceed independently from declined items	 Confirmed
 FR-NEW-17	Aggregated Request status is calculated automatically based on statuses of its RequestItems	 Details TBD (OQ-36)
 FR-NEW-48	[New 16.04] On the "Submit Request" page, a block "Выбрать Work Order из JDE" allows Requestor to browse and select an existing WO from JDE (data sourced from DataLake). The block displays a list of WOs with: number, name, status, priority, date range, and steps with volumes. Upon WO selection, the system automatically generates a draft request pre-populated with WO data. The Requestor then completes the draft by selecting equipment and specifying booking times.	 Confirmed

6.5 Booking Management

ID	Requirement	Status
FR-031	Requestor can add/remove multiple equipment items into a single request; equipment availability in DB is updated	 Confirmed
FR-032	The system prioritizes internal fleet equipments	 Confirmed (as business rule, not technically enforced)
FR-033	Requestor can add internal fleet equipments with "Equipment Usage Status" in (Shared without conditions, Shared with conditions)	 Confirmed
FR-034	The system will suggest external fleet equipments based on a pre-set decision matrix	 Obsolete — replaced by BP showcase model

ID	Requirement	Status
FR-035	Requestor can add external fleet equipment to Request, only if RWA initiator	✗ Obsolete
FR-036	When Requestor adds external fleet equipment to Request, he must provide "Justification"	✗ Obsolete
FR-037	When Requestor adds external fleet equipment, "Cost Center" must be filled	✗ Obsolete
FR-038	Each equipment item added to Request is treated as a separate booking	✓ Confirmed
FR-039	Each booking has its own unique booking ID, start and end datetimes	✓ Confirmed
FR-040	System validates availability of equipment before booking	✓ Confirmed
FR-041	Equipment attributes (picture, fleet, description, usage status, etc.) are displayed	✓ Confirmed
FR-042	Each booking has its own lifecycle (Draft, Submitted, Confirmed, Declined, Revoked, Terminated, Completed)	✓ Confirmed
 FR-NEW-18	Booking start datetime represents the start of mobilization , not arrival at work site	✓ Confirmed
 FR-NEW-19	No mandatory buffer between bookings; equipment availability is validated for the requested period only	✓ Confirmed
 FR-NEW-20	When Requestor adds Unwheeled (Not self-propelled) equipment, the system displays a notification: "Transportation equipment will be added to this booking type"	● Preliminary

6.6 Approval Flow (Internal Fleet Equipment)

ID	Requirement	Status
FR-043	System automatically assigns approver based on fleet ownership	 Confirmed

ID	Requirement	Status
FR-044	If equipment belongs to internal fleet, booking can be confirmed or declined by Fleet Owner (member of corresponding AAD group)	 Confirmed
FR-045	Fleet owner can confirm incoming bookings of equipment in his fleet	 Confirmed
FR-046	Fleet owner can replace the equipment in the booking with another equipment before or after confirmation if booking date is not started or booking end date is in future	 Confirmed
FR-047	Fleet owner cannot replace the equipment if booking is revoked, terminated or booking end date is in past	 Confirmed
FR-048	[Updated 10.04 — CONFLICT-01 RESOLVED] Internal fleet owner can adjust booking date range before or after confirmation ; Requestor is notified automatically; no re-confirmation from Requestor required; equipment availability in DB is updated.	 Confirmed
FR-049	Fleet owner can decline incoming bookings of equipment in his fleet; equipment availability in DB is updated	 Confirmed
FR-050	Fleet owner can indicate 'reason/comment' in case of declining	 Confirmed
 FR-NEW-21	FO must respond to a booking request within a configurable timeout period. Default: 24h — reminder notification; 48h — escalation to manager. Timeout is tracked per RequestItem	 Confirmed
 FR-NEW-22	If FO has free equipment of the requested type but declines, FO must provide a mandatory reason	 Confirmed
 FR-NEW-23	Reminder notification is sent only to the FO of the specific selected equipment item	 Confirmed
 FR-NEW-24	FO presses "Mobilization started" button to record the actual mobilization start time in the system	 Confirmed

6.7 Approval Flow (External Fleet Equipment)

! MAJOR SCOPE CHANGE vs. original BRD (08.04): On-demand BP equipment is **NOT bookable through the system at all**. It is displayed as a showcase/catalog only. All FR-051..057 are **Obsolete**.

ID	Requirement	Status
FR-051	External fleet booking requires CC owner confirmation	 Obsolete
FR-052	System identifies CC owner via PSWS integration	 Obsolete
FR-053	System allows DOAs of CC owners to confirm/decline bookings	 Obsolete
FR-054	System identifies if user is DOA of CC owner via PSWS and DOA integration	 Obsolete
FR-055	CC owner/DOA can put 'reason/comment' when declining	 Obsolete
FR-056	After CC owner confirmation, system requires External FO confirmation	 Obsolete
FR-057	External FO can confirm or decline the booking	 Obsolete
 FR-NEW-25	BP showcase (catalog): Requestor can view on-demand BP equipment by type; each entry shows contract owner and availability condition; no booking action available	 Phase 1
 FR-NEW-26	BP equipment prices/rates are NOT displayed in the system	 Confirmed
 FR-NEW-27	BP populates their own equipment cards in the catalog; contract owner is indicated per entry	 Confirmed
 FR-NEW-28	After FO response timeout with no available internal equipment, Requestor sees a button "Go to BP catalog" in the request	 Confirmed

6.8 Booking Lifecycle Management

ID	Requirement	Status
FR-058	Requestor/SWP can revoke (withdraw) booking if not yet processed by FO; equipment availability updated	✅ Confirmed
FR-059	FO of internal fleet can terminate already confirmed booking; can indicate reason; equipment availability updated	✅ Confirmed
FR-060	Requestor/SWP can terminate already confirmed booking; equipment availability updated	✅ Confirmed (internal fleet only)
FR-061	Requestor/SWP can extend previously confirmed booking; equipment availability updated	✅ Confirmed
 FR-NEW-29	Three-step flow for Unwheeled equipment: (1) Requestor creates request → FO; (2) FO adds transportation equipment → system sends request to "Transportation Responsible" role; (3) Transportation Responsible confirms/declines; (4) FO gives final approval only after transport confirmed	✅ Confirmed
 FR-NEW-30	If transportation equipment is unavailable for Unwheeled request, FO proposes the nearest available date	⚠️ Decision pending
 FR-NEW-31	Booking can be closed early by Requestor or FO via "Close early" button	⚠️ Clarification pending

6.9 Booking/Request Status Management

CONFLICT-02 (FR-070/FR-071 vs. R-75) — ✅ RESOLVED (14.04):

Automatic status transition by datetime is **cancelled**. Booking and Request status changes are **manual only**. FR-070, FR-071, FR-075 → ❌ Obsolete.

ID	Requirement	Status
FR-062	Booking will have 'Submitted' status once Request is submitted	✅ Confirmed

ID	Requirement	Status
FR-063	Booking will have 'Confirmed' status once confirmed by FO	✓ Confirmed
FR-064	Booking will have 'Declined' status once declined by FO	✓ Confirmed
FR-065	Booking of external fleet will have 'Partially Confirmed' status once confirmed by CC Owner	✗ Obsolete
FR-066	Booking of external fleet will have 'Declined' status once declined by CC Owner	✗ Obsolete
FR-067	Status of booking returns to 'Submitted' once extended by Requestor/SWP	✓ Confirmed
FR-068	Booking will have 'Revoked' status once revoked by Requestor/SWP	✓ Confirmed
FR-069	Booking will have 'Terminated' status once terminated by Requestor/SWP/FO	✓ Confirmed
FR-070	Confirmed booking status changes to 'In Progress' once system datetime >= booking datetime	✗ Obsolete (CONFLICT-02 RESOLVED 14.04 — auto-transition cancelled)
FR-071	Confirmed booking status changes to 'Completed' once system datetime < booking datetime	✗ Obsolete (CONFLICT-02 RESOLVED 14.04 — auto-transition cancelled)
FR-072	Request will have 'Draft' status once saved as draft by Requestor/SWP	✓ Confirmed
FR-073	Service Work Request will have 'Draft' status once imported from JDE E1	⚠ Pending (OQ-NEW-C)
FR-074	[Updated 14.04] Request transitions to ' In Progress ' when the first booking within the Request (not counting Declined bookings) transitions to In Progress	✓ Confirmed
FR-075	Request will have 'Completed' status once system datetime >= maximum end datetime	✗ Obsolete (CONFLICT-02 RESOLVED 14.04 — auto-

ID	Requirement	Status
		<i>transition cancelled; see FR-NEW-44)</i>
 FR-NEW-32	Booking closure is manual only: Requestor or FO presses "Close" button. Automatic closure by scheduled end time does NOT apply	✅ Confirmed — supersedes FR-070/071/075
 FR-NEW-33	At the time of closing, Requestor inputs actual start time and actual end time of work; data is used for utilization analytics	✅ Confirmed
 FR-NEW-44	[New 14.04] Request transitions to ' Completed ' when the last active (non-Declined) booking within the Request is closed/completed	✅ Confirmed

6.10 Notifications

ID	Requirement	Status
FR-076	System notifies SWP(s) once Service Work Request is created automatically via API	⚠️ Pending (OQ-NEW-C)
FR-077	System notifies FO once Requestor/SWP submits a Request with internal fleet equipment	✅ Confirmed
FR-078	System notifies FO once Requestor/SWP revokes booking of internal fleet equipment	✅ (presumed confirmed)
FR-079	System notifies Requestor/SWP about results once FO confirms or declines internal fleet booking	✅ (presumed confirmed)
FR-079a	 [Updated 10.04] System notifies Requestor when FO updates the booking period	✅ Confirmed
FR-080	System notifies Requestor/SWP once FO terminates a confirmed internal fleet booking	✅ (presumed confirmed)
FR-081	System notifies FO once Requestor/SWP wants to extend internal fleet booking	✅ (presumed confirmed)

ID	Requirement	Status
FR-082..090	External fleet notification chain (CC Owner, External FO)	✗ Obsolete
 FR-NEW-34	System sends reminder to FO of specific equipment after 24h of no response; escalation to FO's manager after 48h	✓ Confirmed
 FR-NEW-35	System notifies Transportation Responsible role when FO adds transportation equipment to an Unwheeled booking	✓ Confirmed
 FR-NEW-36	System notifies FO when Transportation Responsible confirms or declines transportation readiness	✓ Confirmed
 FR-NEW-43	[New 13.04] System notifications to Fleet Owners are delivered to the fleet's shared team email (and/or to members of the fleet's AD group personal emails); shared email is set per fleet by Admin	✓ Confirmed

6.11 Search & Reporting

ID	Requirement	Status
FR-091	Requestor / SWP can search and filter own requests	✓ Confirmed
FR-092	Approver (CC Owner / FO) can view all pending and completed approvals	✓ Confirmed
FR-093	Admin can view all requests	✓ Confirmed
FR-094	System generates reports on Requests / Bookings / Equipments with capability to apply various filters	✓ Confirmed
FR-095	Admin can view/download reports on all Requests / Bookings / Equipments	✓ Confirmed
FR-096	Approver (CC Owner / FO) can view/download reports on pending and completed Requests / Bookings	✓ Confirmed
FR-097	FO can view/download reports on own equipments	✓ Confirmed
FR-098	SWPs (and selected FOs — Maintenance) can view Service Work Requests grouped in Work Orders	⚠ Pending (OQ-NEW-C)

ID	Requirement	Status
 FR-NEW-37	Audit trail and history of all bookings available on demand (not daily/weekly scheduled reports)	 Confirmed
 FR-NEW-38	Search results for equipment must display as mandatory fields: TCO equipment number + model/make ; license plate (госномер) displayed if present (not all equipment has one)	 Confirmed (<i>Updated 13.04: госномер conditional</i>)
 FR-NEW-39	Search filters in equipment search are dynamic : only parameters relevant to the selected equipment type are displayed; filter matrix (equipment type → filter set) defined in Section 5.1.1	 Partially confirmed (baseline defined; remaining types OQ-NEW-F)
 FR-NEW-45	[New 14.04] System has a dedicated " Completed Requests " page; accessible to Requestor, SWP, FO, Admin with appropriate scope filtering	 Confirmed
 FR-NEW-46	[New 14.04] System generates utilization reports with filters by: date (daily breakdown), department, equipment unit; example output: "actual utilization by day"	 Scope details pending TCO input (FR-094 expansion)
 FR-NEW-47	[Updated 16.04] System generates work center reports (отчёт по рабочим центрам); Work Center = тип работ (шаг в WO). Report groups bookings/requests by Work Center code. (OQ-NEW-G CLOSED)	 Scope details pending

6.12 Dashboards / Map

ID	Requirement	Status
FR-099	System has deployed/integrated visualization tool displaying utilization and coordinates from DataLake/PI (tracker data)	 Confirmed
FR-100	System has deployed/integrated visualization tool displaying digitized location of Work Orders from DataLake (JDE E1)	 Confirmed

 **Action (14.04):** Review and discuss **existing dashboards** at the next meeting (existing TCO dashboard landscape, raw sensor data processing).

6.13 Audit & Compliance

ID	Requirement	Status
FR-101	System logs all request creation, modifications, approvals, rejections	✓ Confirmed
FR-102	Audit trail accessible for compliance purposes	✓ Confirmed
FR-103	Exportable logs for legal/regulatory audits	✓ Confirmed

7 Non-Functional Requirements

Area	Requirement	Status
Availability	99.5% uptime	✓ Confirmed
Performance	The system must support high volume of requests per hour without performance degradation	✓ Confirmed
Security	Role-based access control (RBAC); system integrates with corporate authentication (SSO / Azure AD)	✓ Confirmed
Compliance	Maintain audit history of all approvals / rejections / revokes / terminations	✓ Confirmed
Scalability	Support both internal and external partner use cases	✓ Confirmed
Usability	Intuitive UI for both requesters and approvers	✓ Confirmed
 Users	~200 users (per ACR assessment, 07.04); No PII data stored	✓ Confirmed
 Security tier	ACR assessment result: Low Risk (07.04)	✓ Confirmed
 Authentication	Azure SSO for internal users; separate account flow for external BP users (Phase 2)	✓ Confirmed

Area	Requirement	Status
 UX reference	UI simplicity reference: Booking.com	 Confirmed

8 Integrations

8.1 JDE E1

Work Orders created in JDE E1 must be imported to HDV/HDE tool as Service Work Request(s), once a certain status is assigned to the Work Order. Work Order can contain several Steps; each step is created as a separate Service Work Request in HDV/HDE tool. Only Steps with certain types (BHOE, CDECK, DTRK, FORKL, FTRK-ХИНО, GDRV, GMLOG, MLIFT, VTRK, etc.) will be imported.

The systems must have backward integration which allows changing the status of 'Step' in JDE based on status change of the imported 'Request' in HDV/HDE or by Service Work Processor's manual actions.

 **[Updated 13.04]** Integration path: JDE E1 → **DataLake** (sync every ~30 minutes) → HDV/HDE Booking Tool. Direct JDE API integration is NOT used; data is consumed from DataLake.

 **[Updated 13.04]** JDE Work Order flow: Steps progress: status ~50 (scheduler review by graphist) → status **55** (released to operations). Steps at status 55 are candidates for auto-import as Service Work Request Drafts.

 **[Updated 16.04]** Required Work Order fields from DataLake:

Field	Description / Notes
WO Number	Unique identifier in JDE E1
WO Name	Examples needed from TCO
WO Status	Exact list of statuses in JDE — TBD
WO Status Description	 [New 16.04] Text description of the WO status
WO Priority	Priority value in JDE E1

Field	Description / Notes
Work Center (Step code)	Defines required equipment type; e.g. "CRANE", "DTRK"; FO uses step to identify what equipment is needed
Step Name	Text name of the Work Center step; examples needed from TCO
Step Volumes	 [New 16.04] Required quantity of equipment units per step
Date Range	 [New 16.04] Start date / End date of the WO step

 **[Updated 13.04] Integration contact: Аяжан** — Maintenance Plan Consultant, group TFM 2835 — is the SME for JDE integration and DataLake data structure. Action: schedule meeting to clarify API format, data schema, and exact trigger status.

 **Action (14.04):** Verify available WO fields in DataLake — **Аян, Тельман**.

Status:  Partially confirmed — JDE trigger status TBD; DataLake path confirmed; exact WO field set to be verified in DataLake

 **Repair status (08.04):** Repair status (maintenance in progress / planned completion date) — column "Planned End Date" in JDE — is sourced from DataLake and displayed in Requestor's equipment search. —  Details TBD

8.2 DCT (Digital Contractor Timesheet)

8.2.1 RWA / Change Order Creation

Status:  **Obsolete** — booking of on-demand external equipment is done in DCT directly

8.2.2 Tracker's data receipt

Status:  **Obsolete** — applicable only to external fleet booking workflow

8.3 PSWS (TCO White Page)

Identifies CC owner of cost centers for external fleet booking approval; provides preferred email address for notification recipients.

Status:  **Obsolete** — CC Owner role is not applicable

Note (13.04): White Pages may still be relevant for auto-populating user contact data (email, name) for Fleet Owners. Technical feasibility depends on shared email support in White Pages

API — requires verification (some known bugs with shared emails in previous year). Alternative: manual entry of shared email in Admin Panel (FR-NEW-42).

8.4 DOA (Delegation of Authority)

Identifies if user is DOA of CC owner for external fleet booking processing.

Status:  **Obsolete** — CC Owner/DOA flow is not applicable

8.5 DataLake / PI

From DataLake: utilization information from IVMS and WIALON trackers (extendable to other trackers). Equipment "Tracker ID" field allows system to merge equipment records with DataLake utilization and location data.

Status:  Confirmed

Note (07.04): WIALON integration is **blocked** pending cyber security assessment (Wialon is currently on the restricted list). IVMS and Vega trackers are in scope. — 

 **GIS / Map integration (07.04):** Integration with MAPH/Atlas GIS tool via iframe + URL parameters; equipment ID synchronization; Azure SSO pass-through. —  Pending cyber assessment approval (Светлана)

9 Open Conflicts Summary

#	Conflict	BRD Position	Meeting Decision	Status
CONFLICT-01	FO editing booking period after confirmation	FR-048: editing allowed "before confirmation"	OQ-40: open	 RESOLVED (10.04): FO can edit before AND after confirmation. FR-048 updated.
CONFLICT-02	Booking status auto-transition by datetime	FR-070/071/075: automatic status change by datetime	R-75 (09.04): manual close only	 RESOLVED (14.04): Manual close accepted. FR-

#	Conflict	BRD Position	Meeting Decision	Status
				070, FR-071, FR-075 → Obsolete.
CONFLICT-03	External fleet booking workflow	FR-051..057: full CC Owner → External FO flow	08.04: on-demand BP = showcase only; full booking flow is in DCT	✅ Resolved — FR-051..057 Obsolete

10 Open Questions

OQ ID	Description	Owner	Status
OQ-29	Dynamic filter matrix: equipment type → filter set	TCO	⚠️ Partially defined (see 5.1.1); remaining types → OQ-NEW-F
OQ-30 R-48	One fleet → multiple Fleet Owners: rules and conflicts	—	✅ PARTIALLY CLOSED (13.04) : Fleet has 2+ FOs (shift workers), displayed with shared email. Specific conflict resolution rules TBD
OQ-36	Aggregated Request status mapping (exact state machine)	Team	⚠️ Open
OQ-37	Booking Snapshot — exact field list	Team	⚠️ Open
OQ-39	Split logic	—	✅ Closed
OQ-40	FO editing booking period after confirmation	—	✅ Closed (10.04)
OQ-41	UI details for Unwheeled flow / manual close fields	Team	⚠️ Open

OQ ID	Description	Owner	Status
OQ-NEW-A	Concrete field list for equipment cards per equipment type	TCO Fleet Owners	⚠️ Partially closed (13.04): Loader and Crane-Manipulator defined. Remaining: Generator, Compressor, Dump Truck, Frac Tank, Trailer, Truck — Action Асылбек
OQ-NEW-B	Manage Request / Equipment Template scope — baseline fields first; custom attributes capability → backlog	TCO	⚠️ Open — baseline in progress
OQ-NEW-C	Service Work Processor role definition — is this a user role (graphist/scheduler) or system component? Trigger status in JDE?	TCO (Аяжан, Dias / SA)	⚠️ Open — Action: schedule meeting with Асылбек, Диас (14.04)
OQ-NEW-D	Shared with Conditions vs. Shared without Conditions	—	✅ CLOSED (13.04): Difference = Justification field required. FO recall is independent mechanism.
OQ-NEW-E	Delegation of FO privileges — must be approved by TCO Cybersecurity department before implementation	TCO Cybersecurity / Project Team	⚠️ Open — pending sign-off
OQ-NEW-F	Equipment type parameters not yet covered: Vacuum Truck, and other types from master fleet file	TCO (Асылбек)	⚠️ Open — Action: Асылбек prepares parameter table for all remaining types
OQ-NEW-G	"Work center" (рабочий центр) — is this an attribute of equipment, or a type of work?	—	✅ CLOSED (16.04): Work Center = тип работ (шаг в WO), не атрибут техники. Один тип Equipment → один WC; один WC → несколько типов

OQ ID	Description	Owner	Status
			Equipment. Список кодов — в Glossary Section 3.
OQ-NEW-H	Utilization booking threshold — can a booking be submitted for equipment with utilization approaching 100%? If not, what is the cutoff threshold?	TCO	⚠ Open
OQ-NEW-I	 [New 16.04] Priority level attribution — Is Priority an attribute of: (a) the Request, (b) the Work Center step within the Request, or (c) the equipment item within a step?	TCO + SA	⚠ Open
OQ-NEW-J	 [New 16.04] Department in reports — When generating department-level reports, which department is used: the department of the Fleet Owner or the department of the Equipment itself? (they may differ)	TCO + SA	⚠ Open

11 Requirements Not in Original BRD (New from Meetings)

New ID	Summary	Source	Status
FR-NEW-01	All time parameters configurable via Admin Panel	03.04	✓
FR-NEW-02	FO delegation to any TCO employee with auto-revoke	08.04	⚠ Cybersec pending

New ID	Summary	Source	Status
FR-NEW-03	Admin configures FO timeout thresholds	03.04	✓
FR-NEW-04	Admin authorizes users to book Assigned equipment	08.04	✓
FR-NEW-05	FO uploads multiple photos; Requestor sees photo(s)	08.04, 14.04	✓
FR-NEW-06	Repair status from JDE/DataLake displayed in search and equipment list	08.04, 14.04	⚠
FR-NEW-07	Stationary HDE excluded from Requestor search	08.04	✓
FR-NEW-08	Assigned equipment: visible to all, bookable by authorized only	08.04	✓
FR-NEW-09	Shared with Conditions has visual color marker	03.04	✓
FR-NEW-10	On-demand only; no weekly schedules	03.04	✓
FR-NEW-11	Priority P1–P4 selected by Requestor	03.04	✓
FR-NEW-12	Work Order mandatory; Location replaces it for SCM; WO/WC optional for Logistics	08.04, 16.04	✓

New ID	Summary	Source	Status
FR-NEW-13	"Work Description" mandatory (~50 chars)	08.04	✓
FR-NEW-14	"Comments" optional	08.04	✓
FR-NEW-15	Single form → auto-split into atomic bookings via WO	09.04	✓
FR-NEW-16	Partial confirmation per RequestItem	03.04	✓
FR-NEW-17	Aggregated Request status auto-calculated	09.04 arch	⚠
FR-NEW-18	Booking start = mobilization start	03.04	✓
FR-NEW-19	No mandatory buffer between bookings	09.04	✓
FR-NEW-20	Unwheeled equipment warning notification in UI	07.04	●
FR-NEW-21	FO timeout: 24h reminder / 48h escalation per RequestItem	03.04, 06.04	✓
FR-NEW-22	FO must state reason for decline if equipment available	06.04	✓

New ID	Summary	Source	Status
FR-NEW-23	Reminder to FO of specific equipment only	07.04	✓
FR-NEW-24	FO "Mobilization started" button	09.04	✓
FR-NEW-25	BP showcase: view-only catalog by type	08.04	✓
FR-NEW-26	BP prices not displayed	08.04	✓
FR-NEW-27	BP self-populates catalog cards	08.04	✓
FR-NEW-28	"Go to BP catalog" button after FO timeout	08.04	✓
FR-NEW-29	Three-step Unwheeled flow with Transportation role	09.04	✓
FR-NEW-30	FO proposes nearest date if transport unavailable	03.04	⚠
FR-NEW-31	Early close button for Requestor/FO	07.04	⚠
FR-NEW-32	Manual close only (no auto-transition by datetime)	09.04	✓

New ID	Summary	Source	Status
FR-NEW-33	Actual start/end time entered at close for utilization	09.04	✓
FR-NEW-34	FO reminder 24h / escalation 48h	03.04, 06.04	✓
FR-NEW-35	Notification to Transportation Responsible for Unwheeled	09.04	✓
FR-NEW-36	FO notified of Transportation Responsible decision	09.04	✓
FR-NEW-37	Audit trail on demand	07.04	✓
FR-NEW-38	Search: TCO number + model mandatory; госномер if present	08.04, 13.04	✓
FR-NEW-39	Dynamic search filters by equipment type	08.04	⚠ Partial
FR-NEW-40	Booking calendar in equipment card (FO view)	13.04	✓
FR-NEW-41	Multiple trackers per equipment unit; add trackers from UI	13.04, 14.04	✓
FR-NEW-42	Shared team email per fleet (Admin-managed)	13.04	✓

New ID	Summary	Source	Status
FR-NEW-43	Notifications delivered to fleet shared team email	13.04	✓
FR-NEW-44	Request → Completed when last active (non-Declined) booking closed	14.04	✓
FR-NEW-45	Dedicated "Completed Requests" page	14.04	✓
FR-NEW-46	Utilization reports (by day / department / equipment unit)	14.04	⚠
FR-NEW-47	Work center reports — OQ-NEW-G CLOSED; WC = тип работ	14.04, 16.04	⚠
FR-NEW-48	"Выбрать WO из JDE" block on Submit Request page; WO list from DataLake; auto-generate draft request	16.04	✓
FR-NEW-49	Freeze fields on equipment card: Заморозка, Причина заморозки, Дата завершения заморозки	16.04	✓
FR-NEW-50	Admin manages all reference/handbook values via Admin Panel	16.04	✓